

1 Chapter 1.1

Problem 1. *How many different tickets are possible in each of the following lotteries? And which lottery offers the best chance of winning?*

- (a) *You pick six numbers from 1-16, a number can be picked more than once, and order doesn't matter.*
- (b) *You pick five different numbers from 1-25 and order doesn't matter.*
- (c) *You pick four different numbers from 1-18 and the order in which you specify them matters.*

Answer:

- (a) A multiset combination where $k = 6$ and $n = 16$ which is:

$$\binom{6 + 16 - 1}{6} = \binom{21}{6} = \frac{21!}{6!15!} = 54,264$$

- (b) A subset combination where $k = 5$ and $n = 25$ which is:

$$\binom{25}{5} = \frac{25!}{5!(25-5)!} = \frac{25!}{5!20!} = 53,130$$

- (c) A permutation with no repetition where $k = 4$ and $n = 18$.

$$(18)_4 = \frac{18!}{(18-4)!} = \frac{18!}{14!} = 73,440$$

The second lottery appears to have the best odds of winning.

Problem 2. *You flip a coin 20 times and record the ordered sequence of heads and tails.*

- (a) *How many sequences are there in which you get heads on (at least) flip #1, #4, #7, and #13?*
- (b) *How many sequences have the same number of heads and tails?*

Answer:

- (a) Each choice can be a $\{0,1\}$ for heads or tails. As a result $n = 2$. The positions 1, 4, 7, and 13 have already been decided. This leaves 16 positions available for a head or tails selection. The answer is 2^{16} .
- (b) How many combinations of 20 positions have exactly 10 heads? The answer is $\binom{20}{10}$. How many times can you write a 20 character long string with exactly 10 digits being locked?

Problem 9. *How many subsets of $[20]$...*

- (a) *Have smallest element 4 and largest element 15?*
- (b) *Contain no even numbers?*
- (c) *Have size 10 and don't contain any number larger than 17?*

Answer:

- (a) How many subsets can be written with the numbers between 4 and 15, do not include the empty set. If $B = \{x \in \mathbb{Z} : 4 \leq x \leq 15\} = \{4, 5, 6, \dots, 15\}$. This is asking for the size of the power set of B, but ignoring the empty set. $|\mathcal{P}(B)| = 2^{15-3} = 4096$. The answer is $4096 - 1 = 4095$. This problem can be thought as binary sequence of 12 digits. Each digit corresponds to whether the number 4 - 15 is included. If the digit is a one, it is included. If the digit is zero, it is not included.

- (b) Again, using the analogy that this problem can be thought as a binary sequence with a 0 implying the digit is not included and a 1 implying the digit is included. As such, all the even numbers are automatically swapped to a 0, so only 1 choice. The odd numbers can be a 0 or 1, so two choices. There are 10 odd numbers between 1 and 20. So the answer is 2^{10} .
- (c) This is asking for a subset that is length 10, or $k = 10$. The number of choices has been restricted to 17, so $n = 17$. It is a subset, so no repetition is allowed. As such the answer is $\binom{17}{10}$.

Problem 14. *How many arithmetic problems of the following form are possible? You must use each of the digits 1 through 9, they must appear in numerical order from left to right, and you can use any combination of $+$ and $\times$ symbols you like, as long as the resulting expression makes mathematical sense. For example $1234 + 5 \times 6 \times 78 + 9$ and $123456 + 789$ and 123456789 are three possibilities, but $1 \times \times 234567 + 89$ is not.*

Answer:

In between the sets of two numbers are eight spaces. Each one of these spaces can be inserted with no symbol, a $+$ symbol or a $\times$ symbol. This means there are 3 choices or $n = 3$ for a list of length 8, or $k = 8$. The answer is 3^8 .

Problem 17. *Find the number of 3-lists of the form (x_1, x_2, x_3) , where each x_i is a nonnegative integer and $x_1 + x_2 + x_3 = 10$.*

Answer:

Define $X = \{x_1, x_2, x_3\}$, where X is a multiset. If the values in X are nonnegative and all 3 values must sum to 10, then $X \subseteq A$. Where

$$A = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

Because this is a list, the x_1 sets the appropriate range of the x_2 with $x_2 \leq 10 - x_1$. Then x_1 and x_2 set x_3 with $x_3 = 10 - x_1 - x_2$. As such, only the first two values have multiple options.

1. If $x_1 = 0$ then $x_2 \leq 10$, which has 11 choices.
2. If $x_1 = 1$ then $x_2 \leq 9$, which has 10 choices.
3. If $x_1 = 2$ then $x_2 \leq 8$, which has 9 choices.
4. If $x_1 = 3$ then $x_2 \leq 7$, which has 8 choices.
5. ...
6. If $x_1 = 10$ then $x_2 = 0$, which is 1 choice.

As the value for x_1 increase, the options for x_2 decreases. The answer is represented by $11!$.